

A1 Receive-Coil Amplifier

Hearing-Loop Receiver: Structured Design Review

Daniel Tyukov (5714699)

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Outline

1. Specification & design environment
2. Transfer function & feedback configuration
3. First stage: noise design (g_m , C_{iss} , EKV)
4. Single stage & loop-gain check
5. Dual stage (PMOS pair + NMOS CS)
6. EKV model & pole-zero analysis
7. Frequency compensation (MFM)
8. Conclusions

What the review wants

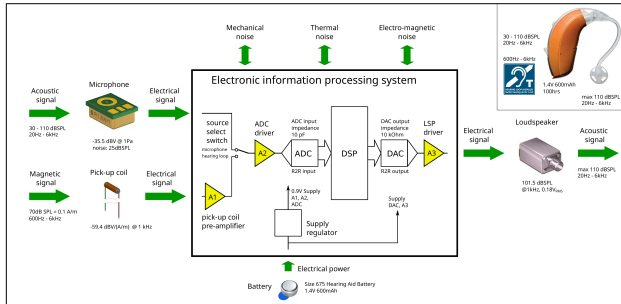
Top-down and *in order*: specify the budget, generate options, then justify every component *before* placing it. Build up to the circuit, don't open with it.

Part 1

Specification & design environment

The hearing loop and where A1 lives

Application diagram



Application diagram, EE4109 HearingLoopIntro

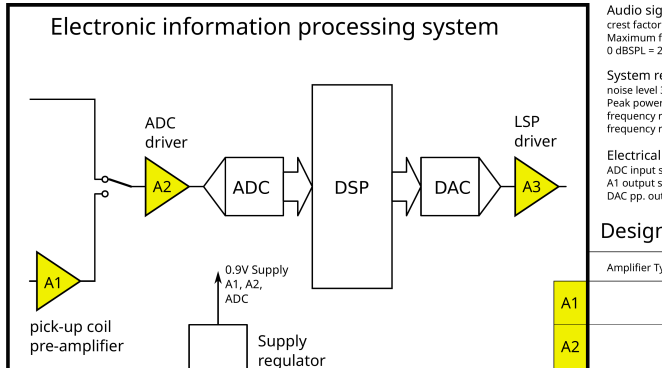
A magnetic *induction loop* broadcasts the audio; a hearing aid picks it up with a coil.

- **A1** = receive-coil pre-amplifier.
- Feeds A2 (ADC driver) → ADC → DSP.

Why

Understand the *environment* first, what feeds A1 and what it drives sets the whole specification.

A1 in the signal chain



A1 is the pick-up-coil pre-amplifier

- **Input:** pick-up coil.
- **Output:** source-select switch (mic / loop), then the ADC driver.
- Single 0.9 V supply; 1 mW power budget.

Three questions first

1. What environment?
2. How does it work?
3. What is the budget?

Input of A1: the pick-up coil

The coil output is proportional to the *rate of change* of magnetic flux, it **differentiates**:

$$v_{coil}(t) = -N \frac{d\Phi}{dt} \propto \frac{dB}{dt}$$

- Current in the loop → magnetic field → coil voltage.
- Open-circuit sensitivity -59.4 dBV/(A/m) at 1 kHz.

Why

If the coil differentiates, A1 must **integrate**, so the cascade is flat across the audio band. That single fact drives the whole topology.

Output of A1: source-select switch + ADC

- **Source-select switch** (mic / hearing loop): needs $V_{mic} = V_{loop}$ at $f_{ref} = 1$ kHz.
- **ADC input** is a voltage amplifier: $Z_{in} \rightarrow \infty$, draws no current.
- Only loading left = parasitic $C_L = 10$ pF .

Consequence

A1 is a **voltage-to-voltage** amplifier driving a capacitor, not a power stage.

Why

Matching $V_{mic} = V_{loop}$ at f_{ref} fixes the required gain magnitude.

How should A1 work?

1. A1 is a **voltage-to-voltage amplifier**.
2. Its transfer is an **integrator** (cancels the coil's differentiation):

$$H(s) = \frac{V_o}{V_i} = \frac{1/\tau_i}{s} = \frac{62.4 \times 10^3}{s} \text{ V/V}, \quad \tau_i = 16.06 \mu\text{s}$$

3. Gain set so $V_{mic} = V_{coil}$ at f_{ref} .
4. **Noise:** weighted output noise $\leq$ microphone floor (≤ 30 dBSPL).

What the review wants

The order matters: transfer first, *then* noise. And it stays a V-V amplifier throughout, no silent switch to a transimpedance amplifier.

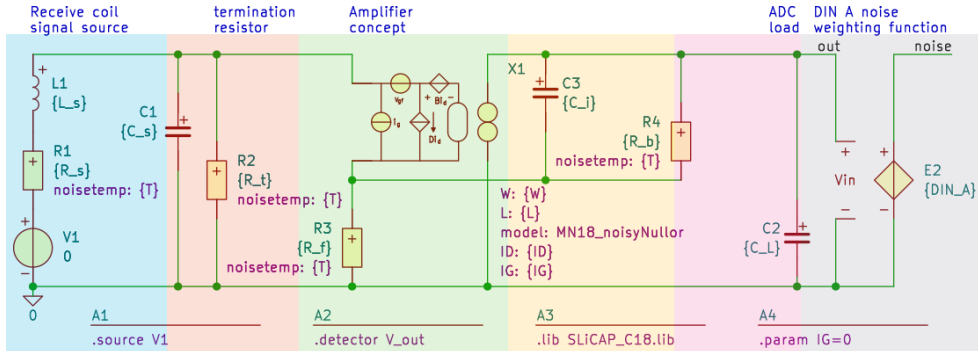
Specification & budget

Quantity	Value	Sets...
Transfer (integrator)	$62.4 \times 10^3 / s$ ($\tau_i = 16.06 \mu s$)	gain / $R_f C_i$
Audio band	$f_{min} = 600 \text{ Hz} \dots f_{max} = 6 \text{ kHz}$	bypass & bandwidth
Output noise (DIN-A)	$\leq 10.6 \mu V$ RMS ($\approx 30 \text{ dB SPL}$)	input stage
Load	$C_L = 10 \text{ pF}$	stability / drive
Supply / power	$1.1 \text{ V to } 1.3 \text{ V} / \leq 1 \text{ mW}$	currents
Coil	$L_s = 120 \text{ mH}, R_s = 875 \Omega, C_s = 9.4 \text{ pF}$	source noise / R_t

Why

Every later decision traces back to one line here. The design is graded on *this reasoning*, not on whether the final numbers are perfect.

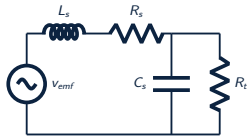
Circuit model: the complete picture



■ coil source ■ termination ■ amplifier (nullor) ■ feedback ■ ADC load ■ noise weighting

The SLiCAP concept model. Each coloured block is represented and justified next.

Representation 1: the pick-up coil



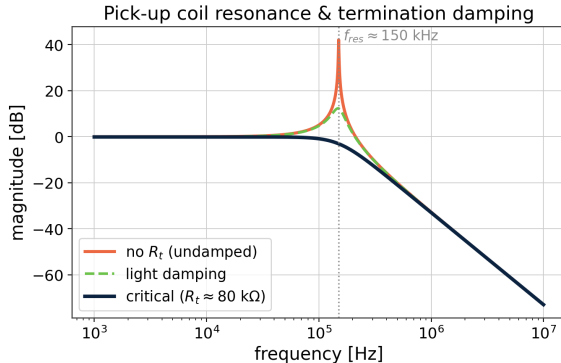
coil source sub-circuit

Parameter	Min	Typ	Max	Unit
DC resistance R_s	788	875	962	Ω
Inductance L_s	102	120	138	mH
Sensitivity @1 kHz	-60.9	-59.4	-57.9	dBV/(A/m)
Resonance freq.		150		kHz

From the quoted resonance, back out the parasitic capacitance:

$$f_{res} = \frac{1}{2\pi\sqrt{L_s C_s}} \Rightarrow C_s = \frac{1}{(2\pi f_{res})^2 L_s} = 9.382 \text{ pF}.$$

Representation 2: the termination resistor



L_s and C_s resonate at 150 kHz; undamped it rings.

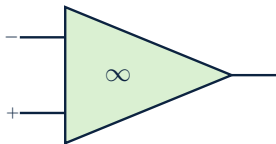
$$Q = 2\pi f_{res} C_s R_t, \quad Q_{crit} = \frac{1}{\sqrt{2}}$$

$$R_t = \frac{Q_{crit}}{2\pi f_{res} C_s} = 79.97 \text{ k}\Omega$$

What the review wants

R_t exists *only* to damp the resonance, and it then becomes a budgeted noise source.

Representation 3: the ideal amplifier (nullor)



nullor: $v_{in} = i_{in} = 0$

Sensitivity gives the coil voltage; with $d/dt \leftrightarrow s$:

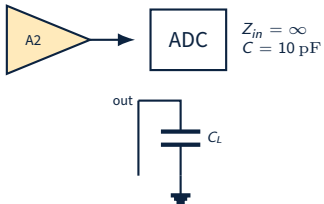
$$V_{ind} = 10^{-59.4/20} = 1.07 \times 10^{-3} \text{ Vm/A}, \quad V_{ind}(s) = k s H(s).$$

Convert to Pa and divide by the microphone sensitivity (-35.5 dBV/Pa):

$$A_{v1} = \frac{V_{mic}}{V_{ind}} = \frac{0.0168}{0.269 \times 10^{-6}} = \frac{62.4 \times 10^3}{s}.$$

With a nullor the transfer is set *only* by the passive feedback, $A_{\infty} = 1/(sR_f C_i)$; the transistors come later.

Representation 4: the output (ADC load)



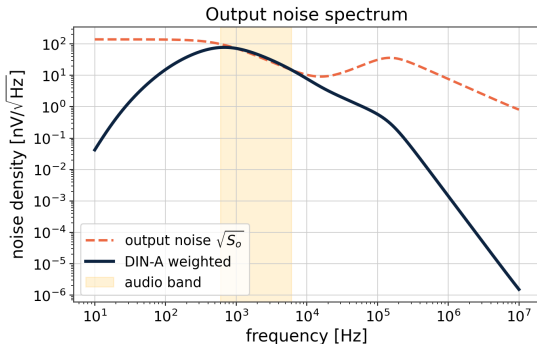
$C_L = 10 \text{ pF}$ is the only load.

The ADC input draws no current ($Z_{in} \rightarrow \infty$); only its parasitic capacitance loads A1.

Why

A capacitive load means the output stage must supply charging current at the full-power frequency, and C_L enters the stability analysis.

Representation 5: the noise sources



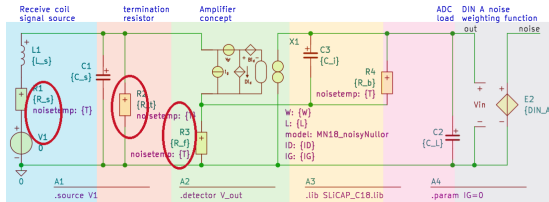
Convert the 30 dB SPL floor to a voltage through the microphone sensitivity ($A_{mic} = -35.5$ dBV/Pa):

$$V_{noise} = 20 \mu V \cdot 10^{(A_{mic} + n_{SPL})/20}$$

$$V_{noise} = 10.62 \mu V$$

Each resistor adds $4kTR$; the input device adds v_n , i_n (the noisy nullor).

Noise budgeting



circled: the three budgeted noise sources R_s , R_t , R_f

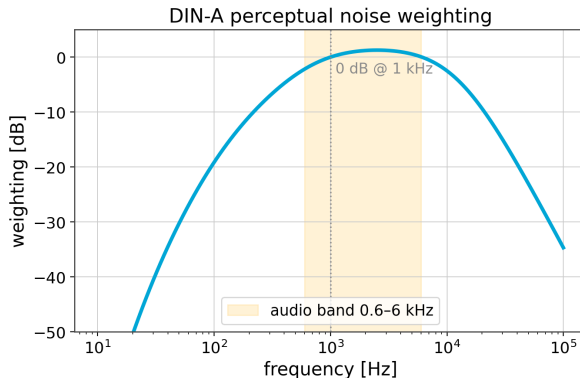
Decide the budget of the *signal source* first:

- Source + termination + feedback: $B_{n1} = 0.4$ of the total.
- Remainder goes to the input transistor, which sets g_m .

$$\text{source share} = \frac{\text{RMS}_{\text{source}}}{V_{\text{onoise}}} \approx 0.14$$

Output-noise RMS $\approx 7.5 \mu\text{V} \leq 10.6 \mu\text{V}$. ✓

Noise is judged by ear: DIN-A weighting



Output noise is weighted by the standard DIN-A curve before integrating:

$$\sigma_o^2 = \int_0^\infty |A(f)|^2 S_o(f) df.$$

- 0 dB at 1 kHz; rolls off outside the audio band.

Why

Weighting makes the budget perceptually meaningful: tight where hearing is sensitive.

Part 2

Transfer function & feedback configuration

First generate the options, *then* choose

Option A, voltage-to-voltage (chosen)

- $Z_1 =$ capacitor, $Z_2 =$ resistor.
- Coil R_s enters only the *noise*.

Option B, current-voltage (transimpedance)

- $Z_2 =$ inductor integrates.
- Coil R_s enters the *transfer*; large inductor.

What the review wants

The prereview lesson: **pick one and stay consistent**. Mixing a V-V amplifier with a transimpedance amplifier between slides makes the plots meaningless.

Decision

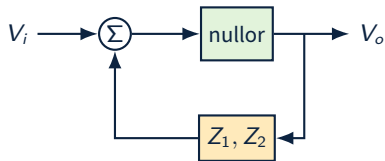
Option A: voltage-to-voltage integrator. Practical components, and R_s stays out of the transfer.

Why feedback? The asymptotic-gain model

Feedback fixes the transfer with passive components, so accuracy follows them, not the transistor.

$$A_f = A_\infty \underbrace{\frac{-L}{1-L}}_{\text{servo}} + \underbrace{\frac{1}{1-L}}_{\text{direct}}$$

- A_∞ : ideal (nullor) transfer = $1/(sR_f C_i)$.
- L : loop gain; servo $\rightarrow 1$ when $|L| \gg 1 \Rightarrow A_f \rightarrow A_\infty$.



What the review wants

In the review, *explain every curve* of this model, gain, asymptotic, loop gain, servo, direct.

Sizing the feedback network

Bound R_f from two budgets:

- **Noise** ($B_{n1} = 0.4$) $\Rightarrow$ upper bound $R_{f,max} = 7.5 \text{ k}\Omega$.
- **Power** ($B_{i1} = 0.1$) $\Rightarrow$ lower bound $R_{f,min} = 1.4 \text{ k}\Omega$.

$$C_i = \frac{\tau_i}{R_f}, \quad R_b = \frac{1}{2\pi C_i f_{min}}.$$

$$R_f = 3.24 \text{ k}\Omega$$

$$C_i = 4.96 \text{ nF} \quad R_b = 53.5 \text{ k}\Omega$$

Why

$R_b \parallel C_i$ sets the low corner at $f_{min} = 600 \text{ Hz}$ and gives the DC bias a path, one resistor, two justified jobs.

Part 3

First stage: noise design

The first stage sets the input-referred noise

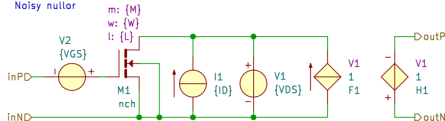
NMOS noisy nullor circuit

Models a nullor with NMOS input-referred noise sources
 NMOS parameters: W, L, M, ID, VGS, VDS (ID>0, VGS>0, VDS>0)
 Put the library directive in the top-level circuit

A1

```
.subckt noisyNullorN outP outN inP inN W=220n L=180n M=1 ID=1u VGS=0.6 VDS=0.9
```

Noisy nullor



Noisy nullor: ideal nullor + the input device's v_n , i_n

Model the controller as a **noisy nullor**.

- Later stages' noise is divided by the first-stage gain $\Rightarrow$ negligible.
- So the whole budget is spent on *stage 1*.

Why

This is why the first stage is designed *from the noise spec*, before anything else.

The knobs: g_m , c_{iss} , inversion

Input-referred thermal noise of a CS stage:

$$\overline{v_n^2} = \frac{4kT\gamma}{g_m}$$

- More $g_m \Rightarrow$ less noise (more current for less voltage).
- Wider device $\Rightarrow$ more $c_{iss} \Rightarrow$ more input loading.
- **Weak inversion** gives the largest g_m/I_D .

The trade-off

Minimise $g_m \cdot c_{iss}$: enough g_m for the noise budget, lowest c_{iss} for the source.

What the review wants

“Replace the how with the why”: weak inversion is chosen *because* it maximises transconductance efficiency.

Which stage? CS vs CG vs CD

	Gain	Chain (ABCD)	Role
CS	high	smallest entries	most <i>nullor-like</i>
CG	≈ 1 (current)	—	current follower
CD	≈ 1 (voltage)	—	voltage follower

- The CS chain matrix has the smallest elements $\Rightarrow$ most nullor-like, so later stages barely add noise.
- CG / CD are followers, no voltage gain to suppress later noise.

First stage = **common-source**, in saturation.

From budget to device parameters

1. Find the output noise spectrum $S_o(f)$.
2. Integrate (DIN-A weighted) $\rightarrow$ variance $\sigma_o^2 = (B_{n2} V_{o,n})^2$.
3. Express g_m as a function of c_{iss} .
4. Lowest feasible $c_{iss} \rightarrow g_m, I_D$, then W, L via the **EKV** model.

PMOS input ($g_m \cdot c_{iss}$ optimum):

$$g_m = 73 \mu\text{S}, W = 19.1 \mu\text{m}$$

$$L = 0.72 \mu\text{m}, I_D = 8.1 \mu\text{A}$$

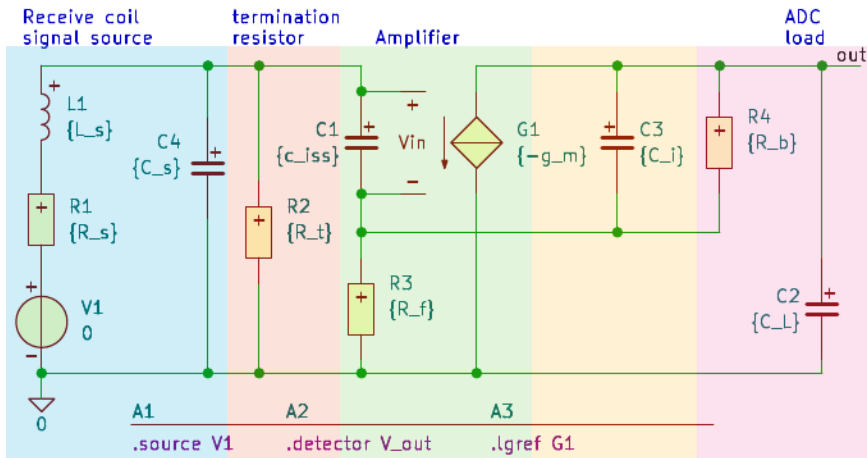
Why

SLiCAP gives the symbolic spectrum; EKV turns the budget into W, L, I_D ; ngspice verifies.

Part 4

Single stage & loop-gain check

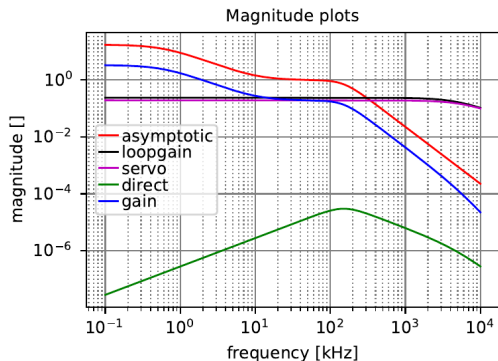
Single stage: the circuit



■ coil source ■ termination ■ amplifier ($-g_m$) ■ feedback ■ ADC load

behavioural model: one common-source controller, i.e. the single amplifier stage

Is a single stage enough?



Loop gain L stays below 0 dB.

A single CS has only 3 usable terminals $\Rightarrow$ a **differential pair** ($\approx 4 \times$ current/area).

- Current drive into C_L : ✓
- Voltage headroom on 0.9 V: ✓
- Loop gain $|L_{DC}| < 1$ across the band: ✗

$|L| < 1 \Rightarrow$ servo cannot enforce the transfer $\Rightarrow$ **poor accuracy. Go to a dual stage.**

Part 5

Dual stage (PMOS pair + NMOS CS)

Why a dual stage?

Five competing costs: *noise, power, accuracy, stability, area.*

- The **first stage dominates noise**.
- The **second stage dominates loop gain** (accuracy).

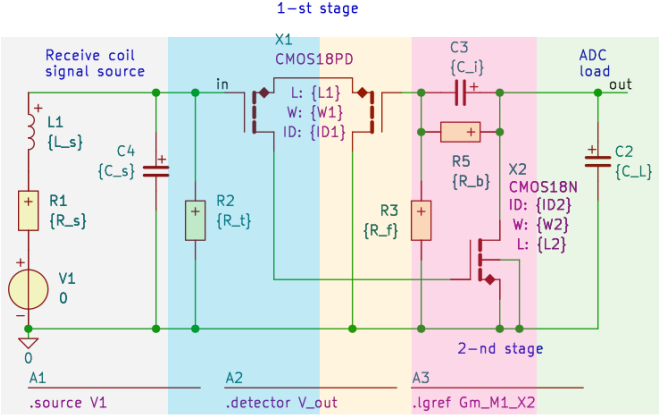
Split the job, optimise each stage for one thing.

Stage 1: **PMOS differential pair**, low noise.

Stage 2: **NMOS common-source**, high loop gain.

Why

One stage could not give both low noise and enough loop gain; two stages can.



■ coil ■ 1st stage: PMOS pair ■ feedback ■ 2nd stage: NMOS CS ■ ADC load

EKV schematic, .source V1, .detector V_out, .lgref Gm_M1_X2

First stage: differential pair vs single CS

Differential pair

- + lower noise
- + high CMRR
- + matches the coil's 2-terminal input
- more area
- higher complexity

Single-ended CS

- + less area
- + lower power
- + simple
- allows CM interference
- higher noise

The coil is a floating two-terminal source $\Rightarrow$ the **differential pair** wins on noise and common-mode rejection.

Second stage: NMOS common-source

Chosen for what the first stage is *not* optimised for:

- high loop gain $\Rightarrow$ accuracy;
- high current-drive into C_L ;
- low power, small area, simple.

Minimum length for speed; width traded for g_m/I_D .

NMOS CS output stage:

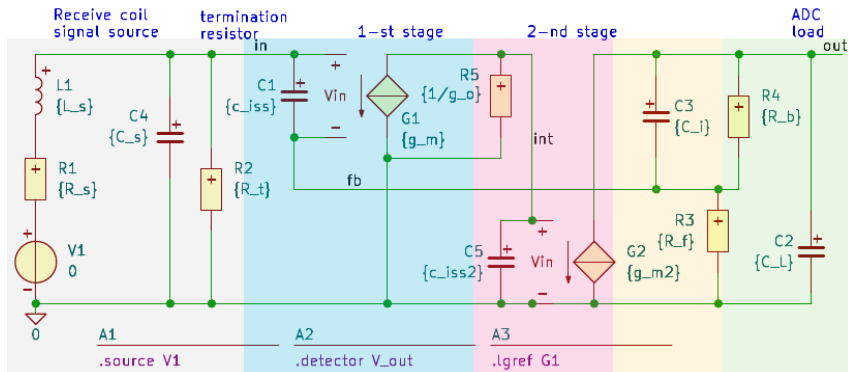
$$W = 10 \mu\text{m}, L = 0.18 \mu\text{m}$$

$$I_D = 16.5 \mu\text{A}$$

Why

Drive current $I_D \approx V_{i,pp}/2R_f$ must charge C_L at the full-power frequency.

Dual-stage design parameters



■ coil ■ 1st stage (g_{m1} , g_{o1}) ■ 2nd stage (g_{m2}) ■ feedback (R_f) ■ ADC load

g_{m1} first-stage transconductance, set by the *noise* budget.

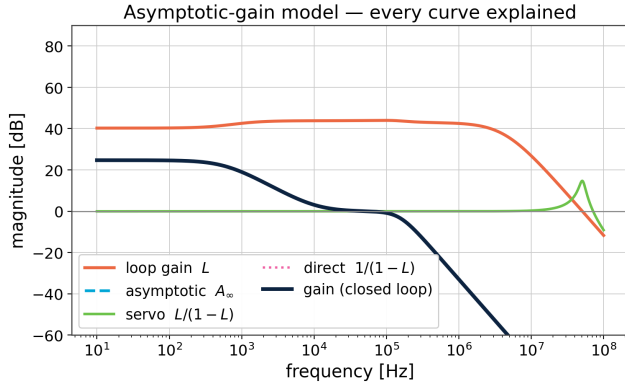
g_{m2} second-stage transconductance, the extra *loop-gain* factor.

R_f feedback resistor, set by the *transfer*.

g_{o1} first-stage output conductance (its output resistance).

$$L_{DC} = \frac{R_f g_{m1} g_{m2} (-R_s - R_t)}{R_s g_{o1} + R_t g_{o1}} = -\frac{R_f g_{m1} g_{m2}}{g_{o1}}$$

Asymptotic-gain model: every curve explained



- gain**: the closed-loop transfer we deliver.
- asymptotic** A_∞ : ideal nullor target $1/sR_f C_f$.
- loop gain** L : ≈ 45 dB at DC, budget for accuracy.
- servo** $L/(1-L)$: $\rightarrow 1$ while $|L| \gg 1$, so gain $\rightarrow A_\infty$.
- direct** $1/(1-L)$: feed-through; negligible.

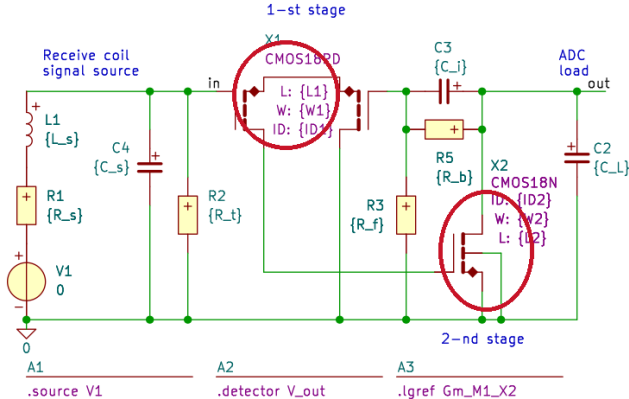
What the review wants

Gain follows A_∞ exactly where the servo ≈ 1 ; it departs once $|L|$ falls.

Part 6

EKV model & pole-zero analysis

EKV small-signal model



same topology as slide 32; the two **transistors** (circled) are now full EKV models

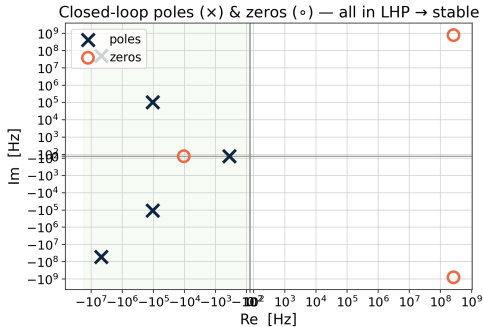
The circled devices carry what the ideal nullor and behavioural model left out:

- finite output conductance g_o ;
- parasitic caps C_{gs} , C_{gd} , C_{gb} , C_{db} , C_{sb} .

Why

These parasitics are what create the high-frequency poles & zeros, so now the stability check is trustworthy.

Pole-zero analysis



	Re [Hz]	Im [Hz]	$ f $ [Hz]	Q
p_1	-602	0	602	—
$p_{2,3}$	-106.5k	$\pm 106.6k$	150.7k	0.71
$p_{4,5}$	-4.69M	$\pm 50.9M$	51.1M	5.44
$z_{1,2}$	+263M	$\pm 794M$	837M	1.59
z_3	-10.5k	0	10.5k	—

- All poles $\text{Re} < 0 \Rightarrow$ **stable**.
- The 51 MHz pair has $Q \approx 5.4$, severely underdamped $\Rightarrow$ peaking.
- The gain plot follows the poles & zeros, as expected.

Part 7

Frequency compensation (MFM)

Target: maximally-flat magnitude (MFM)

A pole pair with $Q > \frac{1}{\sqrt{2}}$ peaks and rings. **MFM** (Butterworth) sets the dominant pair to

$$Q = \frac{1}{\sqrt{2}} \approx 0.707.$$

- No peaking, fastest flat response.
- Comfortable phase margin.

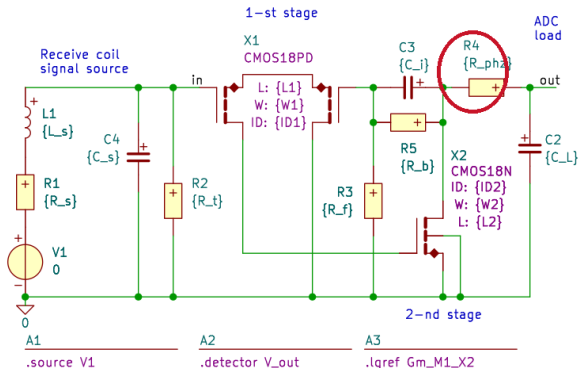
How to move a pole pair?

Add a zero to the loop gain near the offending poles, it injects phase lead and lowers the effective Q .

Why

We shape *stability*, not just gain, a Bode plot is the 0-axis slice of the Nyquist plot.

The phantom-zero method



actual SLiCAP circuit; the one added component, R_{phz} , is circled

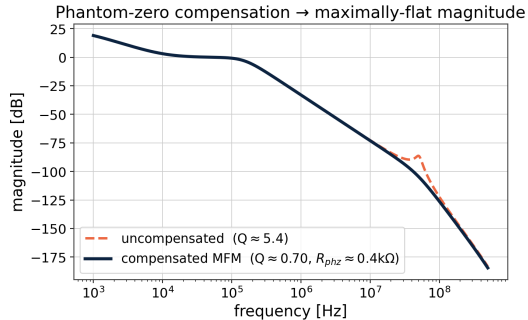
Put R_{phz} in series with the load: a zero in the *loop gain*, with no real pole in the signal path (the feedback is tapped *before* it).

$$z = -\frac{1}{R_{phz} C_L}$$

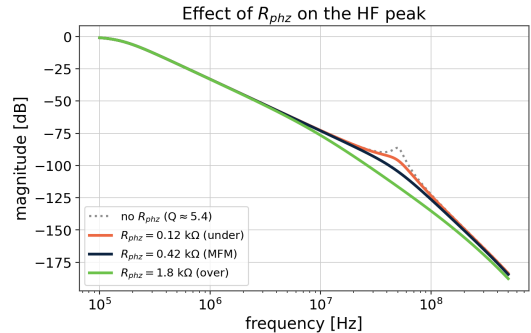
Why this one?

- no extra noise source in the signal path;
- negligible power and area;
- no compensation capacitor (no added energy storage).

Compensation result



full band: peak removed, passband intact



tuning R_{phz} : under / MFM / over

$R_{phz} \approx 0.42 \text{ k}\Omega$ tunes the 51 MHz pair Q : 5.4 → 0.70 (MFM); all poles stay in the LHP, so it remains stable. Cost: one resistor.

Part 8

Conclusions

Conclusions

The structured path, end to end:

1. Specification & budget from the environment.
2. V-V integrator chosen *after* listing options.
3. Build the model up, block by block.
4. Noise budget $\rightarrow$ first-stage g_m , $C_{iss} \rightarrow W, L, I_D$.
5. Single stage fails accuracy $\rightarrow$ dual stage.
6. PMOS pair (noise) + NMOS CS (loop gain).
7. EKV pole-zero: stable, one high- Q pair.
8. Phantom zero $\rightarrow$ maximally-flat magnitude.

What the review wants

The grade is in the *reasoning*: right order, options before choices, every curve and component justified, the how replaced by the why.

$$H(s) = \frac{62.4 \times 10^3}{s}, \text{ noise } \leq 10.6 \mu\text{V, stable, MFM.}$$

Thank you

Questions & discussion